

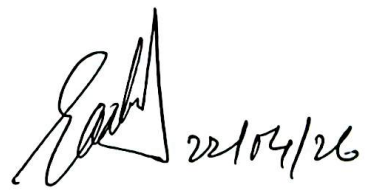


DAFFODIL INTERNATIONAL UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FYDP (TITLE PHASE) EVALUATION REPORT

REPORTING PERIOD- SPRING 2026

Project Identification:

I. Project Title	Transparent Predictions: Using Interpretable Dynamic Model to Understand University Student Dropout Factors	
II. Group Members	ID: 0242310005101361	Name: Arnob Das
	ID: 0242310005101583	Name: Md. Shazzadur Rahman Piash
III. Supervisor	Name: Mr. Saiful Islam	Designation: Assistant Professor
IV. Co-Supervisor	Name: Md Tanvirul Islam	Designation: Lecturer
V. Submission Date:	22 April, 2026	
VII. Certificate:	“This is to certify that the final year design project work until Title defense evaluation held on _____, titled as stated in Sec. I, executed by the students group mentioned in Sec. II, have been found satisfactory and every section of this report is reflecting the same.”	 (Signature of Supervisor & date)

CO Description and Corresponding PO for Title Defense Phase

CO	CO Descriptions	PO
CO1	Integrate recently gained and previously acquired knowledge to identify a real-life complex engineering problem for the “Transparent Predictions: Using Interpretable Dynamic Model to Understand University Student Dropout Factors” project.	PO1
CO2	Analyze different aspects of the goals in designing an engineering problem solution for the “Transparent Predictions: Using Interpretable Dynamic Model to Understand University Student Dropout Factors” project.	PO2
CO3	Explore diverse problem domains through a literature review, delineate the issues, and establish the goals for the FYDP..	PO4

1. Introduction

University student dropout remains one of the most pressing challenges in higher education globally. This issue significantly impacts students’ personal and professional futures, as well as institutional performance and societal development. Despite its severity, existing dropout prediction systems fail to explain the reasoning behind their predictions, leaving educators unable to design effective and targeted interventions.

This project proposes the development of a Transparent Predictions System — an interpretable, dynamic machine learning model designed to identify university students at risk of dropping out while providing clear, explainable insights for each prediction. By integrating Long Short-Term Memory (LSTM) for temporal modeling and SHAP (Shapley Additive explanations) for interpretability, this system aims to bridge the gap between predictive accuracy and explainability. The ultimate goal is to empower academic advisors and institutional authorities with actionable, evidence-based tools to intervene early and effectively in student retention efforts.

2. Background Study

Student dropout prediction has emerged as a significant area of research as universities seek to improve retention rates and support student success. Researchers have applied various artificial intelligence and machine learning techniques to identify at-risk students, with studies spanning multiple countries and educational contexts.

Early research used statistical methods like logistic regression for interpretability. Over time, random forests gained popularity for their higher accuracy. More recently, neural networks have shown great promise, though they raise concerns about interpretability and transparency.

Key findings from literature highlight several important patterns. Academic performance, especially first-semester grades and credits earned, consistently predicts dropout risk. Attendance patterns strongly correlate with withdrawal decisions. Financial factors including scholarships, loans, and tuition

payments also play a significant role. Demographic characteristics such as age at enrollment, gender, and family background provide additional predictive power.

However, most existing models suffer from critical limitations. The majority provide only static end-of-semester predictions, limiting the timing of possible interventions. The most accurate models function as black boxes, offering no insight into their reasoning. Furthermore, research does not reveal how specific thresholds — such as the number of absences that trigger high risk — affect dropout likelihood. The interaction effects between various factors also remain poorly understood in the existing literature.

3. Problem Statement

The core problem this project addresses is the lack of transparency and timeliness in existing university dropout prediction models. Current systems predict student dropout risk but fail to explain the reasons behind those predictions. As a result, academic advisors and administrators receive at-risk lists without understanding whether the underlying problems stem from poor attendance, declining academic performance, or financial difficulties.

Without meaningful explanations, interventions become guesswork rather than informed actions. Additionally, existing static models only provide predictions at semester-end, which significantly limits the window for early intervention. Students who could be supported in time are often identified too late for effective remediation.

This project aims to solve these problems by developing a system that not only predicts dropout risk dynamically over time but also clearly explains why each student is flagged — and what specific thresholds in attendance, grades, or finances are driving that risk.

4. Preliminary Novelty Statement

This project introduces a novel combination of temporal deep learning and post-hoc explainability techniques to address a widely recognized but underserved problem in educational data mining. While individual components such as LSTM and SHAP have been applied in various domains, their integrated use for dynamic and interpretable student dropout prediction represents a fresh approach.

The key novelties include: (1) real-time risk updates using LSTM that tracks student behavior across multiple time steps rather than relying on end-of-semester snapshots; (2) per-student explanations generated by SHAP, revealing exactly which factors are driving each individual's risk; (3) threshold identification that exposes critical tipping points — for example, at what number of absences risk sharply escalates; and (4) integration of pre-admission data to enable earlier and more proactive identification of at-risk students.

5. Research Questions

The central research question guiding this project is:

- 1. How effectively can a dynamic temporal model (LSTM) predict student dropout compared to traditional machine learning models?**
- 2. How can explainable AI techniques (SHAP) identify key factors and critical thresholds influencing student dropout?**

6. Objectives

The key objectives of this project are:

- Develop a dynamic, temporal dropout prediction model using Long Short-Term Memory (LSTM) that continuously updates risk scores based on evolving student data.
- Integrate SHAP (Shapley Additive Explanations) to provide transparent, per-student explanations of why each individual is flagged as at-risk.
- Incorporate pre-admission records, demographic data, financial status, and real-time academic progress to enable comprehensive and early risk detection.
- Identify critical risk thresholds — such as the exact number of absences or grade drops — at which dropout probability sharply increases.
- Create actionable, personalized risk reports for academic advisors, enabling targeted and evidence-based student interventions.
- Compare the performance of the proposed dynamic model with existing static models to quantify the added value of temporal modeling.

7. Expected Outcome

Upon successful completion, this project is expected to deliver the following outcomes:

- A functional and interpretable LSTM-based dropout prediction system that dynamically updates risk scores as new student data becomes available throughout the semester.
- SHAP-based explanation reports for individual students, clearly indicating the most influential factors driving their dropout risk.
- Discovery of critical behavioral and academic thresholds — quantifying, for example, how many consecutive absences or failed courses trigger high-risk classification.
- Quantification of the predictive value of pre-admission data, demonstrating whether early indicators meaningfully improve risk identification.
- Empirical evidence comparing LSTM-based dynamic predictions against static traditional models, demonstrating the advantage of temporal approaches.
- A methodology that can be adopted or adapted by other universities for personalized student support systems.

8. Scope of Work

This project is bounded by the following scope and limitations:

- The study will focus on undergraduate students at universities; postgraduate or part-time students are excluded from the scope.
- Student data to be used includes pre-admission records, demographic information, financial records, and academic performance data (first and second semester). Only anonymized data collected with appropriate ethical approvals will be used.

- The primary modeling technique will be LSTM for temporal dynamics, supplemented by baseline models (logistic regression, random forest) for comparative analysis. SHAP will be used for interpretability.
- The model will be evaluated on dropout prediction at the end of the academic year. Real-time deployment and live integration with university systems are outside the scope of this project.
- The system will not prescribe specific interventions but will provide risk explanations and threshold information to support human decision-making by academic advisors.
- Class imbalance in the dataset will be addressed using SMOTE; however, dataset size and availability may limit the generalizability of findings across diverse institutions.

9. References

- [1] L.-S. Chen, T.-T. Huynh-Cam, V. Nalluri, T.-C. Lu, and S. Agrawal, "Determining Important Features for First-Year Student Dropouts Using Artificial Intelligence Algorithms," in 2024 6th International Workshop on Artificial Intelligence and Education (WAIE), 2024, pp. 128–133, doi: 10.1109/WAIE63876.2024.00030.
- [2] M. Vaarma and H. Li, "Predicting student dropouts with machine learning: An empirical study in Finnish higher education", Tech. in Soc., 2024, vol. 76, pp. 102474.
- [3] N. Mduma, "Data balancing techniques for predicting student dropout using machine learning", Data, 2023, vol. 8, pp. 49.
- [4] J. D. Febro, "Utilizing feature selection in identifying predicting factors of student retention", Inter. J. Adv. Com. Sci. and App., 2019, vol. 10.
- [5] P. Berka and L. Marek, "Bachelor's degree student dropouts: Who tend to stay and who tend to leave?", Stu. Edu. Eval., 2021, vol. 70, pp. 100999.

Supervisor's Feedback: Provide comments and suggestions for improvement

1. Title needs attention
2. Background study is not ok
- 3.

Examiner's Signature: _____

Date: _____